

CVE-2020-17144 : Microsoft Exchange Server EWS Insecure Deserialization

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Content

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CVE-2020-17144 : Microsoft Exchange Server EWS Insecure Deserialization

With the recent events, there have been speculations [1] regarding increasing cybersecurity attacks on organizations by the threat actors. Amidst the speculations, name of a CVE which has popped up is CVE-2020-17144 which is a Remote Code Execution (RCE) vulnerability in Microsoft Exchange Server 2010 SP3.

What is Microsoft Exchange

Microsoft Exchange is a mail and calendaring server implemented using ASP.NET. Microsoft Exchange Server provides web access for Exchange Web Services (EWS) which is an application program interface (API) that allows programmers to access Microsoft Exchange items such as calendars, contacts, and email. Vulnerability Details

This CVE affects all the versions of Microsoft Exchange 2010 SP3 till Cumulative Update (CU) 30 (released on 08.12.2020). There is a Proof of Concept (PoC) publicly available on Github [2]. We will test out the PoC on a test machine in a moment. Before we do that, here is a brief, simplified explanation of the root cause of the vulnerability -

EWS uses **SOAP** (Simple Object Access Protocol) messages, which are XML based [3] to access and modify the user configuration object. One of the methods

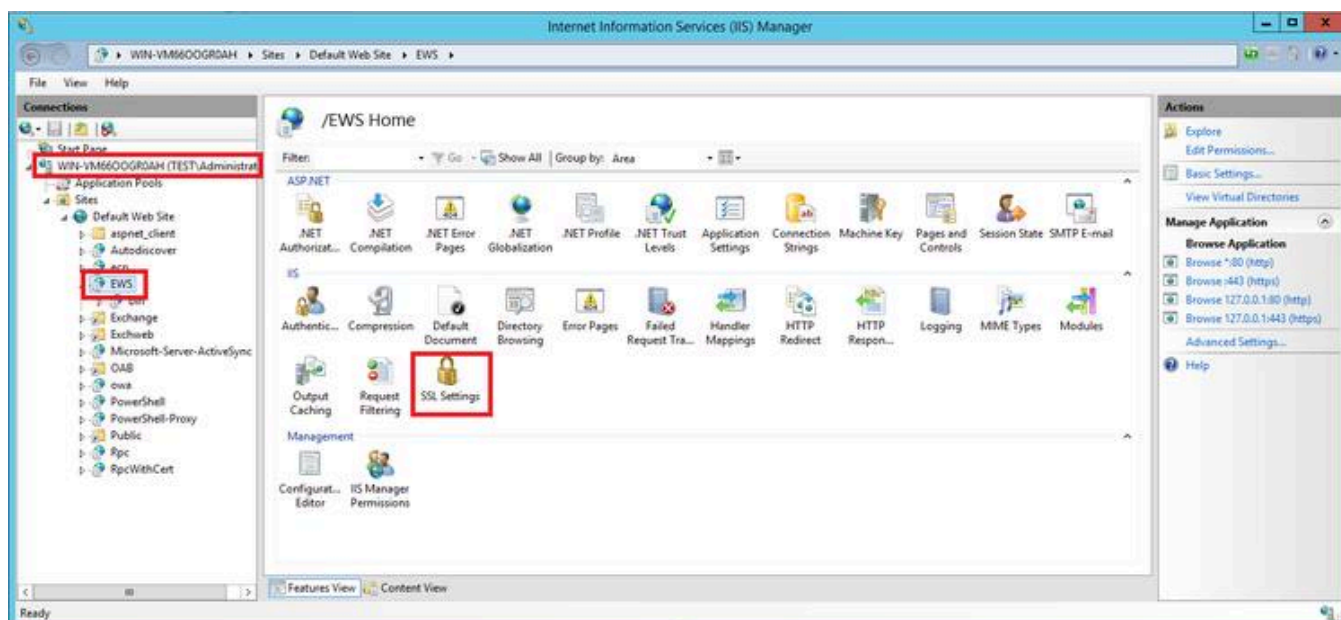
(**CreateUserConfiguration**) accepting SOAP messages has a field called “binary data” which accepts serialized [4] data. This serialized data is de-serialized on the server side without any validation. So, an authenticated attacker can embed malicious data in these SOAP messages capable of performing RCE causing the vulnerability.

With that out of the way, let's see the attack in action.

Attack in action

Configuring the target

- Firstly, we need a Windows OS which supports deployment of Microsoft Exchange Server. For that we deploy a Windows Server 2012 VM. We can get one evaluation ISO from here [5].
- We will use the same VM to launch the attack too. We can use some other machine on the same network as well.
- Active Directory (AD) Domain needs to be deployed on this machine. We can follow the guide from one of my previous blogs [6] in the ‘Deploying the Domain Controller’ section.
- After deploying an AD, we need to install the Microsoft Exchange Server 2010 SP3 [7].
- The version downloaded won't have any Cumulative Update installed. Visual Studio C++13 [8] is needed if we want to apply the latest vulnerable Cumulative Update 30 [9] over it.
- Next, we need to deploy the MS Exchange server 2010, we can follow the tutorial here [10].
- Once the Exchange setup is done, we need to make sure we can login to the following URLs by visiting them in the browser - **https://localhost/owa** and **https://localhost/EWS/exchngex.asmx**
- Optional: We can disable SSL for EWS to see what's being exchanged in the wire in plain text. To monitor we need to download a network monitoring tool like Wireshark [11].
- To disable SSL, search for IIS manager and go to the SSL settings of EWS and remove the check on the Require SSL option.



Preparing the payload

- We download the Public PoC available online [2] and navigate to it inside Command Prompt
- The payload needs to be compiled, but before compiling, if SSL has been disabled for EWS, open the CVE-2020-17144.cs file in a text editor and replace the word https to http.
- Now in command prompt, run the batch script called make. This should create the CVE-2020-17144 file which is our payload.

Executing the Payload

To give an idea of what the payload is designed to do is, it creates a serialized object, which hosts a malicious server/webpage on the server, which can respond to user query by executing the commands that is supplied, whoami in this case.

Now, to launch the attack, run,

- **CVE-2020-17144 <ip_address> <username> <password>**
- Replace the IP address, username, and password with corresponding details
- This should return successfully with message 'User Configuration Created', which means the attack probably has been successful.

To test if the attack was successful -

- Visit **http://10.39.44.241/ews/soap/?pass=whoami** in the browser
- This should return the username which means command execution was successful

Associated Strikes

<Enter Search Criteria> Advanced ▾

Displaying 1 of 1 | [Get more results](#)

StrikeId	Name	Prot...	Dire...	Referen...	Sev...	Varl...
E22-9dfc1	Microsoft Exchange Server EWS UserConfiguration Insecure Deserialization	http	c2s	CVE: 20...	MED...	1

Name:
Microsoft Exchange Server EWS UserConfiguration Insecure Deserialization

Path:
/strikes/exploits/misc/cve_2020_17144_microsoft_exchange_server_ews_userconfiguration_deserialization.xml

Description:
This strike exploits a remote code execution vulnerability in Microsoft Exchange Server 2010. The vulnerability is due to improper handling of malicious EWS requests. An authenticated, remote attacker can exploit the vulnerability by sending crafted requests to the vulnerable Exchange server. Successful exploitation could result in the remote code execution in the context of the System user.

Category:
Exploits: Miscellaneous

Reference(s):
Type: CWE Value: 502
Type: CVE Value: 2020-17144
Type: URL Value: <https://github.com/zcgonvh/CVE-2020-17144>
Type: URL Value: http://www.zcgonvh.com/post/analysis_of_CVE-2020-17144_and_to_jeapionizing.html
Type: CVSSv3 Value: 8.4 CVSS:3.1/AV:N/AC:L/PR:H/UI:R/S:C/C:H/I:H/A:H
Type: CVSS Value: 6.0 (AV:N/AC:M/Au:S/C:P/I:P/A:P)

Keyword(s):
network_service

Customers of BreakingPoint now have access to attack campaigns for different advanced persistent threats, allowing them to test their currently deployed security controls' ability to detect or block such attacks. For more details, see [BreakingPoint](#).

References

- <https://www.marketscreener.com/quote/stock/RAPID7-INC-23055722/news/Rapid7-Stayin-g-Secure-in-a-Global-Cyber-Conflict-39579278/>
- <https://github.com/zcgonvh/CVE-2020-17144>
- https://www.w3schools.com/xml/xml_soap.asp
- <https://devopedia.org/data-serialization>
- <https://www.microsoft.com/en-in/evalcenter/evaluate-windows-server-2012-r2>
- /blogs/tech/nwvs.entry.html/2020/12/15/hp_data_protector-jA0g.html
- <https://www.microsoft.com/en-in/download/details.aspx?id=36768>
- <https://www.microsoft.com/en-in/download/details.aspx?id=40784>
- <https://www.microsoft.com/en-us/download/details.aspx?id=100910>
- <https://www.youtube.com/watch?v=kPhABk20bac>
- <https://www.wireshark.org/download.html>
- <https://support.ixiacom.com/version/breakingpoint-strikepacks-os-v830-0/>

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